MIL-E-27675B(USAF)
10 August 1964
SUPERSEDING
MIL-E-27675A(USAF)
17 September 1962

MILITARY SPECIFICATION

ENGINE PRESSURE RATIO SYSTEM, TURBOFAN ENGINE A/A24J-13

1. SCOPE

1.1 This specification covers one type of turbofan engine pressure ratio measuring system with vertical tape scale presentation, designated as A/A24J-13.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-C-675	Coating of Glass Optical Elements (Anti-Reflection)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5501	Caps and Plugs, Protective, Dust and Moisture Seal
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery

FSC 6620

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:41Z
Check the source to verify that this is the current version before use.

MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-20708	Synchros, 60 and 400 Cycle, General Specification for
MIL-M-26512	Maintainability Requirements for Aerospace Systems and Equipment
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-701	Preferred and Guidance List of Transistors
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-831	Test Reports, Preparation of
MS27034	Connector, Receptacle, Electrical, Pin Insert, Cylindrical, Miniature, Hermetic, Solder Mount
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33586	Metals, Definition of Dissimilar

Air Force-Navy Aeronautical

AND10050	Bosses, Standard Dimensions for Gasket Seal Straight Thread

(Copies of documents required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The system furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein and has been listed on or approved for listing on the applicable qualified products list.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:41Z
Check the source to verify that this is the current version before use.

3.2 Units. The Engine Pressure Ratio System, Turbofan Engine A/A24J-13 shall consist of:

Item Nr	Qty	Item Name	Reference Paragraph
1	4	Transmitter, Pressure TRK-55/A24J-13	3.7.1
2	2	Indicator, Pressure EGK-7/A24J-13	3.7.2

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials and parts

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent to be approved by the procuring activity.

3.4.2 Metals. Metals shall be corrosion-resistant or suitably treated to resist corrosion due to conditions likely to be encountered in storage or normal service.

3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.2.2 Magnesium alloys. Magnesium alloy parts shall not be used.

3.4.3 Electronic component parts. Electronic component parts and the application thereof shall conform to MIL-E-5400. Transistors shall be utilized and shall be selected in accordance with MIL-STD-701. Vacuum tubes shall not be used. The control transformers required for the servos in the indicator shall be compatible with the output synchro transmitters specified for the EPR transmitters.

3.5 Design and construction. The Engine Pressure Ratio System, Turbofan Engine A/A24J-13 shall be designed to measure and indicate engine pressure ratio. One transmitter shall be used in conjunction with each engine. Each transmitter shall drive the two indicator channels (hooked in parallel), one for the pilot's panel and one for the flight engineer's panel.

3.5.1 Reliability. System reliability shall be a major design criterion. The system shall be so designed and constructed as to provide maximum practicable periods of time-between-failures. Failure shall be defined as any malfunction that results in degraded performance, regardless of the cause (careless handling or involuntary destruction excepted).

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T12:41Z
Check the source to verify that this is the current version before use.

3.5.1.1 The transmitter and indicator shall have a minimum mean-time-between-failures (MTBF) value of at least 750 hours based on the demonstration tests outlined in section 4 herein. The design goal of the system shall be 2000 hours MTBF based on service experience.

3.5.2 Maintainability. The system shall be so designed and constructed that organizational and field maintenance shall be accomplished by removing a known defective transmitter or indicator and replacing same with a workable item. The system shall be easily adjusted and required by personnel of depot bases using depot level repair and test equipment. Maintainability shall be in accordance with MIL-M-26512.

3.5.3 Electrical requirements.

3.5.3.1 Power consumption. The maximum power consumption shall not exceed 50 va for a single transmitter and its respective two indicator tape displays.

3.5.3.2 Voltage. The system shall operate from a 115V, 400 cycle, single-phase power source in accordance with MIL-STD-704. The synchro transmitters and control transformers required for the system shall operate from the same electrical power source and shall fulfill the functional requirements specified in MIL-S-20708.

3.5.4 Construction. The system shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service.

3.5.4.1 Pivots, bearings, and gears. Pivots, bearings, and gears shall not bind or shake, and shall be as frictionless as practicable. Gears shall have antifriction backlash provisions where required for accuracy.

3.5.4.2 Screw threads. Screw threads shall be in accordance with MIL-S-7742.

3.6 Performance. The engine pressure ratio system shall receive engine air intake total pressures (P_t2) and exhaust total pressures (P_t7) and process them into an indicated engine pressure ratio reading used indirectly to determine thrust.

3.6.1 TRK-55/A24J-13 transmitter. Each transmitter for each engine shall sense the compressor inlet total pressure (P_t2), sense the turbine outlet total pressure (P_t7), compute the ratio by dividing P_t7 by P_t2, and provide this synchro output signal to the indicators.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:41Z
Check the source to verify that this is the current version before use.

3.6.1.1 Response time. The transmitter response time shall be within the limits specified in table I.

TABLE I. Response Rate

Engine Inlet Pressure (P_t2) (Inches Hg Absolute)	Engine Exhaust Pressure (P_t7) (Inches Hg Absolute)		Ratio Change	Allowable Time (Seconds)
3.50	4.55	4.90	0.10	0.5
3.50	4.55	8.75	1.20	5.0
3.50	7.00	7.70	0.20	1.0
15.00	19.50	37.5	1.20	5.0
15.00	33.00	37.5	0.30	1.5
28.00	36.40	70.00	1.20	5.0
40.00	52.00	80.00	0.70	3.5
42.00	54.60	92.40	0.90	4.5

3.6.1.2 Sensitivity. The change in pressure necessary to cause a change of indication shall not exceed the values specified in table II.

TABLE II. Sensitivity Test Pressures

Inlet Pressure P_t2 (Inches Hg Absolute)	Exhaust Pressure P_t7 (Inches Hg Absolute)	Pressure Ratio	Sensitivity (In. Hg)	
			P_t2 Constant P_t7 Varying	P_t2 Varying P_t7 Constant
3.00	5.40	1.80	0.008	0.008
11.00	22.00	2.00	0.030	0.030
42.00	75.60	1.80	0.120	0.120

3.6.1.3 Pressure ranges. The compressor inlet total pressure (P_t2) shall vary from 1.5 to 85 inches Hg absolute and the turbine outlet total pressure (P_t7) shall vary from 2 to 150 inches Hg absolute. The differential pressure (P_t7 - P_t2) shall range from 0 to 140 inches Hg. A maximum overpressure value of 10 percent shall be applied to the P_t2 and P_t7 ranges.

3.6.1.4 Temperatures. The transmitter shall withstand the following ambient temperature ranges:

a. -54° to 100° C, operational to specified tolerances

b. -65° to 121° C, exposure and operation.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:41Z
Check the source to verify that this is the current version before use.

3.6.1.5 Endurance. Tha transmitter shall withstand an endurance test of at least 1726 hours while undergoing cycling operation from approximately 1.0 engine pressure ratio to 2.7 engine pressure ratio and return at a rate of not less than 3 cpm, as outlined in section 4 herein.

3.6.2 EGK-7/A24J-13 indicator. The two indicators shall simultaneously receive single synchro output signals from four separate transmitters as specified herein, convert these signals to four servo-driven vertical tape indications, and display separately and remotely the pressure ratios of the four respective turbofan engines, within specified accuracies.

3.6.2.1 Decompression. The indicator shall withstand explosive decompression of the surrounding atmosphere. Operation shall not be affected when a pressure differential of 16 inches Hg is applied instantaneously between the interior and exterior of the case.

3.6.2.2 External pressure. The sealed indicator assembly shall withstand an external pressure of 56 inches Hg absolute without failure of the cover glass and hermetic seal and without case distortion that would affect normal operation.

3.6.2.3 Temperature. The indicator shall withstand the following ambient temperature ranges:

a. -54° to 71° C, operational to specified tolerances

b. -65° to 71° C, exposure and operation.

3.6.2.4 Magnetic effect. When held in any of the positions specified in section 4 herein, the indicator shall not cause deflection of the compass by more than 5°.

3.6.2.5 Independent operation. The indicator shall be so designed that no failure in any portion of the instrument for one engine will cause failure, malfunction, or erroneous indication in any portion of the instrument for another engine.

3.6.2.6 Endurance. The indicator shall withstand an endurance test of at least 1726 hours while undergoing 12,000 cycles of operation from approximately 1.0 engine pressure ratio to 2.3 engine pressure ratio and return at a rate of not less than 3 cpm, as outlined in section 4 herein.

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:41Z
Check the source to verify that this is the current version before use.

3.6.3 Dielectric strength

3.6.3.1 Nonhermetically sealed components. Insulation for electrical circuitry and parts that carry 65 to 115 volts and are housed in nonhermetically sealed cases shall be sufficient to withstand a potential of 1,250 volts ac rms between the circuitry and the case. Insulation for electrical circuitry and parts that carry less than 65 volts shall be sufficient to withstand a potential of 500 volts ac rms between the circuitry and the case.

3.6.3.2 Hermetically sealed components. Insulation for electrical circuitry and parts that are housed within hermetically sealed cases shall be sufficient to withstand a potential of 200 volts ac rms between the circuitry and the case. However, this insulation shall meet the requirements for insulation within nonhermetically sealed cases, in the event that the cases are unsealed to disconnect sensitive parts and tests are performed with the cases unsealed.

3.7 Details of units

3.7.1 Transmitter, pressure TRK-55/A24J-13

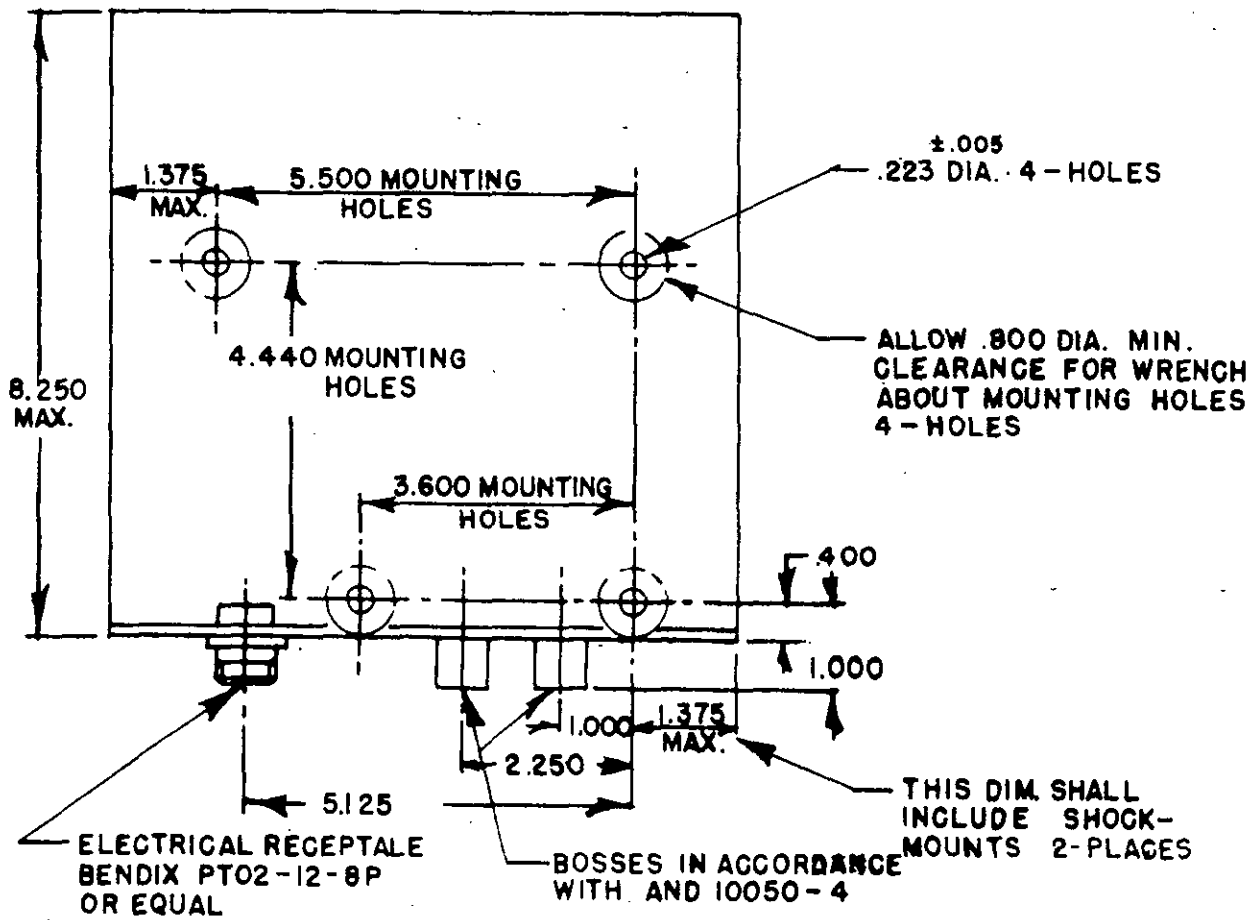
3.7.1.1 Size and weight. The outline dimensions of the transmitter shall not exceed those shown on figure 1. The weight of the transmitter shall be kept to a minimum and shall not exceed 5.5 pounds.

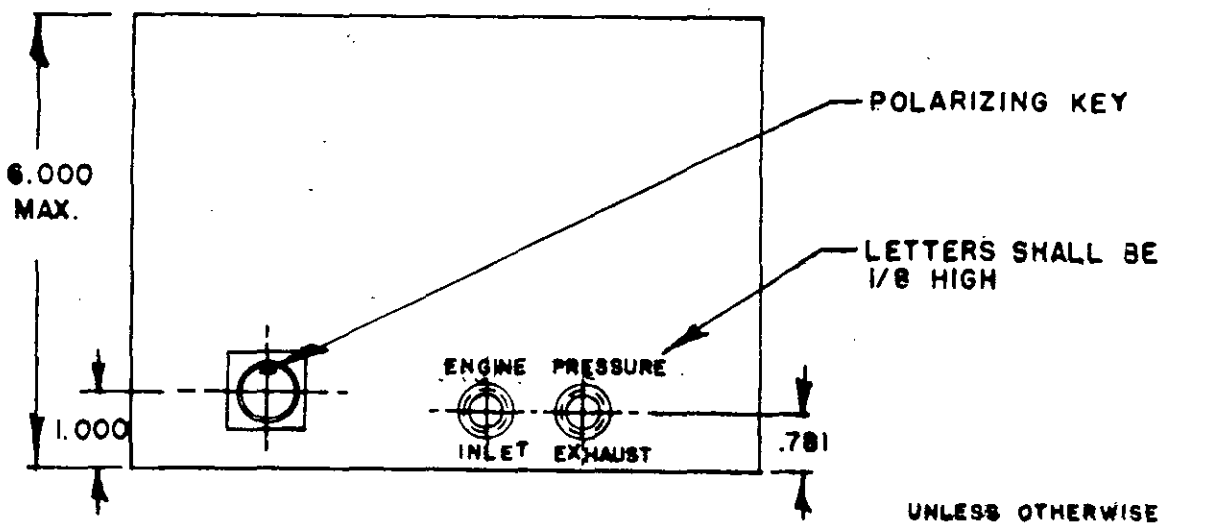
3.7.1.2 Case. The case for housing the components of the transmitter shall be fabricated of aluminum. It shall be designed to provide the degree of ruggedness and sealing characterictics required for the transmitter to satisfactorily fulfill the applicable tests specified in section 4. It shall be so designed that the case may be opened, component parts removed and replaced, and the case resealed without the use of special tools. Hermetic sealing of the case is optional.

3.7.1.3 Fittings. The P_t7 and P_t2 pressure connection fittings and location thereof shall be as shown on figure 1. Metal tubing shall be used to connect rack fittings to transmitter fittings, if required.

3.7.1.3.1 Protective closures. The pressure connections shall be provided with protective closures in accordance with MIL-C-5501 to protect the threads and sealing surfaces and prevent the entrance of foreign material.

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:41Z
Check the source to verify that this is the current version before use.





UNLESS OTHERWISE SPECIFIED DIMENSIONS IN INCHES. TOLERANCES:
FRACTIONS ± 1/64
DECIMALS ± .010

FIGURE 1 TRANSMITTER OUTLINE

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:41Z
Check the source to verify that this is the current version before use.

3.7.1.4 Electrical receptacle. The electrical receptacle shall be Bendix type PT02-12-8P, or equal. The receptacle shall be wired as follows:

PIN	CONNECTION
A	Power ground side.
B	Power input 115V, 400 cps.
C	S_3 of synchro transmitter stator.
D	S_1 of synchro transmitter stator.
E	S_2 of synchro transmitter stator.
F	Case ground
G	Spare
H	Spare

3.7.1.5 Stops. The transmitter mechanism shall include suitable stops that will allow the indicator to register beyond either end of the scale. The upper limit stop shall be set between 2.700 and 2.745 EPR and the lower limit stop shall be set between 0.955 and 1.000 EPR.

3.7.1.6 Screw safety-lock. The transmitter external body screws and internal screws shall be secured with safety wire or suitable lock washers. The method selected shall be approved by the procuring activity prior to use.

3.7.1.7 Mounting. A mounting rack with vibration isolators shall be provided and shall be fabricated of aluminum. Mounting holes and the location thereof shall be as shown on figure 1.

3.7.1.8 Damping. Damping shall be incorporated in the transmitter design to prevent excessive oscillation of the output signal. The degree of damping shall be consistent with applicable test requirements specified in section 4.

3.7.1.9 Calibration. The transmitter shall be so calibrated that the rotor of the output synchro transmitter rotates over a range of 323° for a corresponding pressure ratio of 1.0 to 2.7 sensed by the transmitter. The electrical zero setting of the output synchro transmitter shall be such that the transmitter will satisfactorily operate one channel of the indicator as specified in section 4.

3.7.1.9.1 Electrical zero setting. The electrical zero setting of the output synchro transmitter shall be determined as follows:

a. Connect the primary and secondary as shown on figure 2, procedure 1. The synchro shaft will rotate to either of two positions that are 180° apart.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:41Z
Check the source to verify that this is the current version before use.

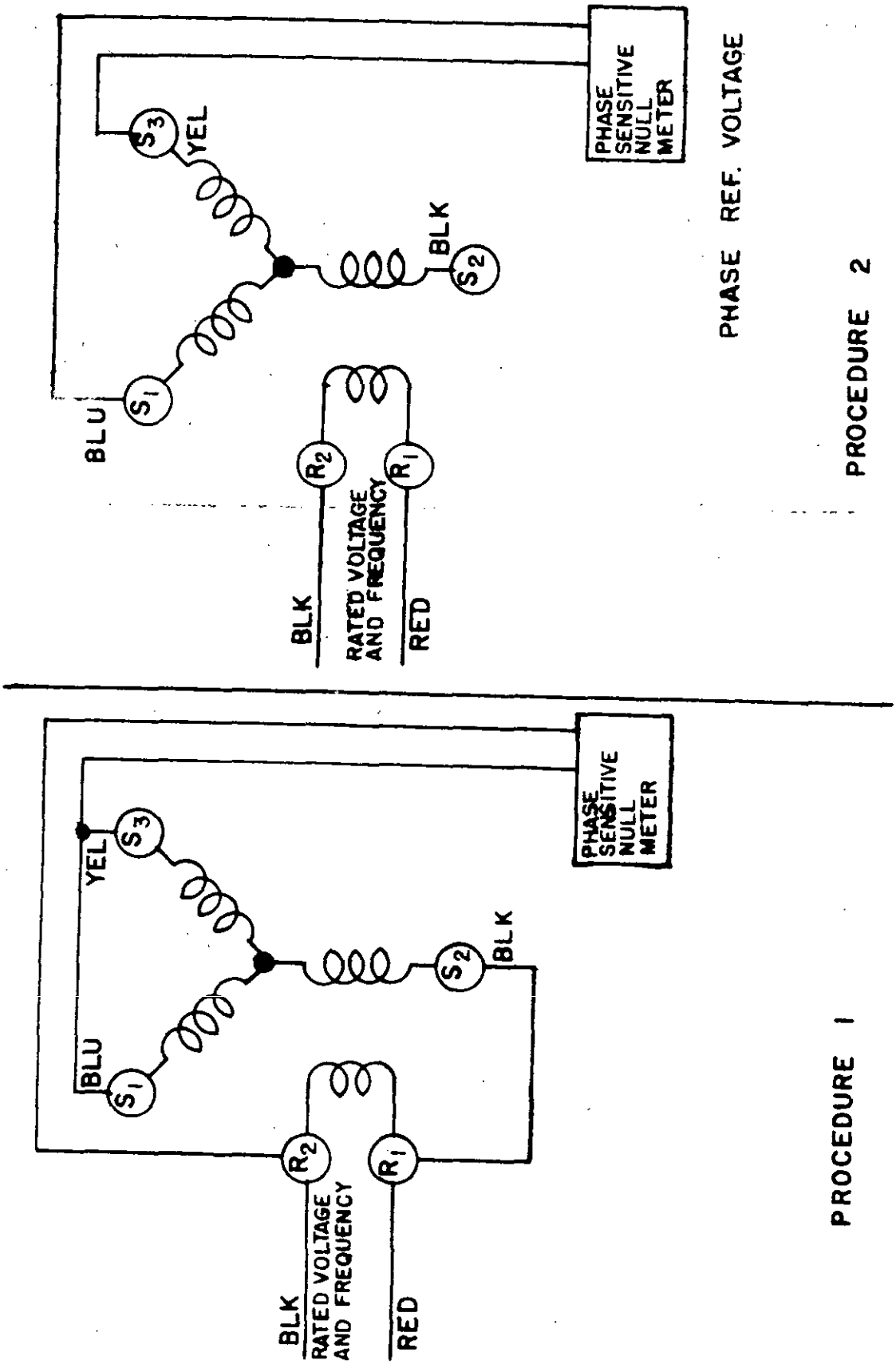


FIGURE 2 ELECTRICAL ZERO SETTING

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:41Z
Check the source to verify that this is the current version before use.

The correct approximate zero position is that position which produces the smaller voltage reading in the phase sensitive null meter.

b. To accurately determine the exact synchro zero position, reconnect the synchro as shown on figure 2, procedure 2 without rotating the synchro shaft. Then rotate the synchro shaft through the smaller angle that will produce a minimum reading on the phase sensitive null meter. When the synchro shaft is rotated in a counterclockwise direction from electrical zero as viewed from the rotor shaft end, the voltage $E(S_1\ S_3)$ shall be increasing and in approximate time phase with $E(R_2\ R_1)$. This position is the synchro electrical zero position.

When increasing pressure ratio values are sensed by the transmitter, the output signals produced shall cause the respective indicator to likewise register compatible increasing readings. The same relationship shall apply for decreasing pressure ratio measurements. An increasing pressure ratio indication shall correspond to a clockwise rotation of the synchro transmitter as viewed from the rotor shaft end when installed in the pressure ratio transmitter. Lead R_2 of the transmitter synchro shall correspond to power input ground when installed in the pressure ratio transmitter.

3.7.1.9.2 Internal wiring. Where practicable, the internal wiring shall be color coded.

3.7.1.10 Fail-safe provision. The transmitter shall be so designed that if a circuit fails within the transmitter, causing the servo motor to fail to operate but leaves the output synchro operational, the output synchro shall return to the 1.0 EPR stop position. This shall result in the indicator tape moving to the 1.0 EPR position for the particular channel affected. Compliance with this requirement shall be demonstrated by a detailed fault analysis as specified in section 4 herein.

3.7.2 Indicator, pressure EGK-7/A24J-13

3.7.2.1 Size and weight. The outline dimensions of the indicator shall not exceed those shown on figure 3. The weight of the indicator shall not exceed 5 pounds.

3.7.2.2 Indicator case. The indicator case shall be designed in accordance with figure 3 and shall be constructed of a material that is uniform in texture with a smooth external surface and suitable for hermetic sealing. The hermetic seal shall be impervious to any atmospheric condition that may be encountered by the indicator. The case shall be so designed that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools and fixtures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:41Z
Check the source to verify that this is the current version before use.

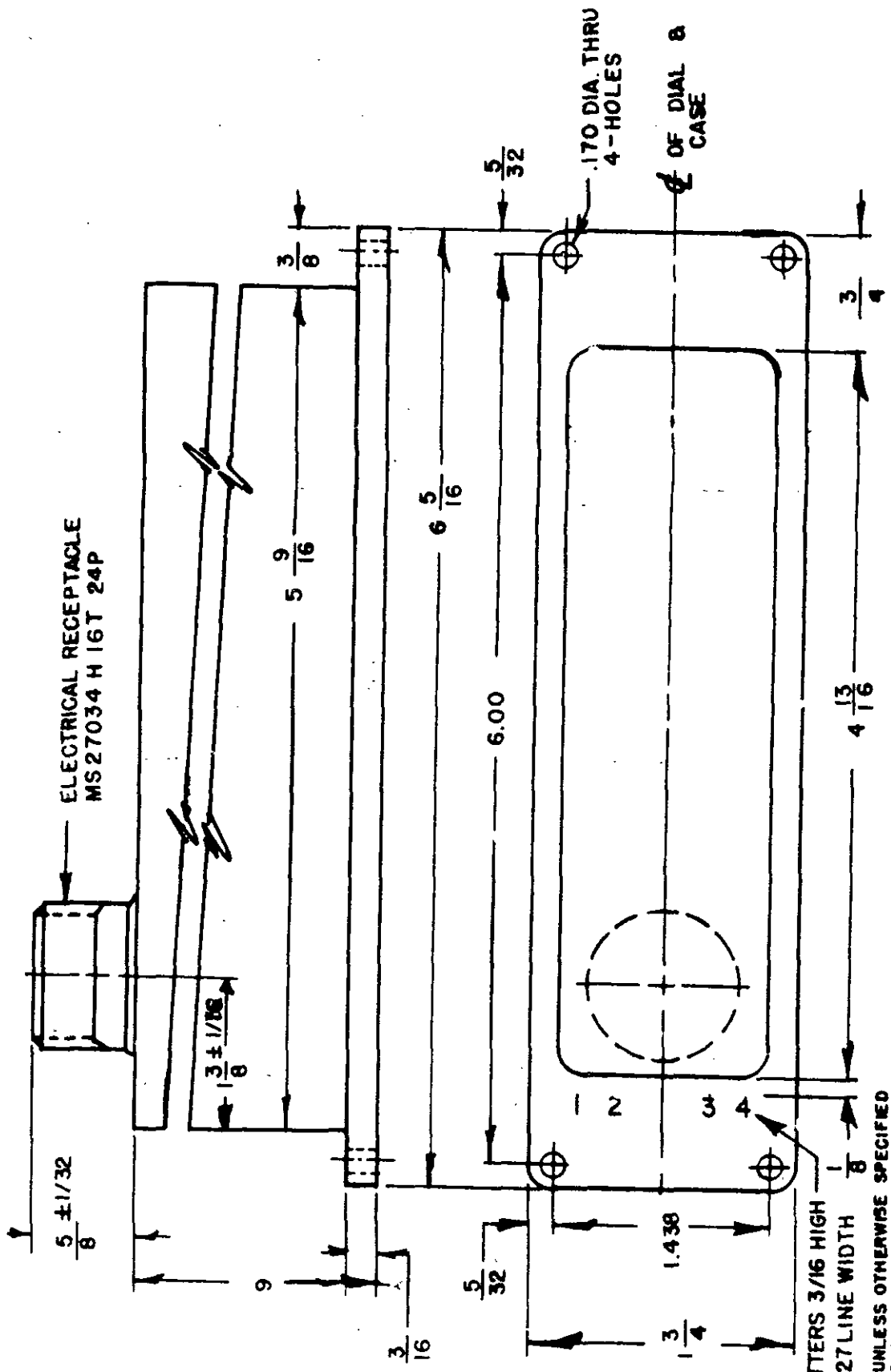


FIGURE 3 INDICATOR CASE OUTLINE

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:41Z
Check the source to verify that this is the current version before use.

3.7.2.2.1 Filling medium. The case shall be filled with 100 percent helium, or a mixture of 88 to 92 percent nitrogen and the remainder helium, or 95 percent helium and 5 percent carbon dioxide. The filling medium shall be of at least 98 percent purity, free of duct particles, and shall contain not more than 0.006 mg of water vapor per liter (dew point -65° C) at the filling pressure. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere.

3.7.2.2.2 Filling tube. If used, the filling tube shall be at least 1-1/4 inches long and shall not be visible when the indicator is completely assembled.

3.7.2.3 Electrical receptacle. The indicator shall be provided with an MS27034-H-16T-24P electrical receptacle. A means shall be provided for safety wiring the interconnecting wiring connectors to the indicator case and shall be subject to the approval of the procuring activity.

3.7.2.4 Indicator mechanism. The indicator mechanism shall be so sufficiently rugged that small distortions of the case will not affect the accuracy or operation of the mechanism.

3.7.2.4.1 The amplifiers required for the servo systems in the indicator shall be contained within the indicator.

3.7.2.4.2 Each synchro receiving device in the indicator shall be so designed that when its rotor position is 180° or more away from the null position commanded by the transmitter synchro signal, the receiving device shall automatically sense this condition and drive itself to the correct position relationship with the transmitter within 6 seconds after application of excitation power to the indicator. If a null seeking device is used, actuation of such device shall occur at a pressure ratio below 0.950.

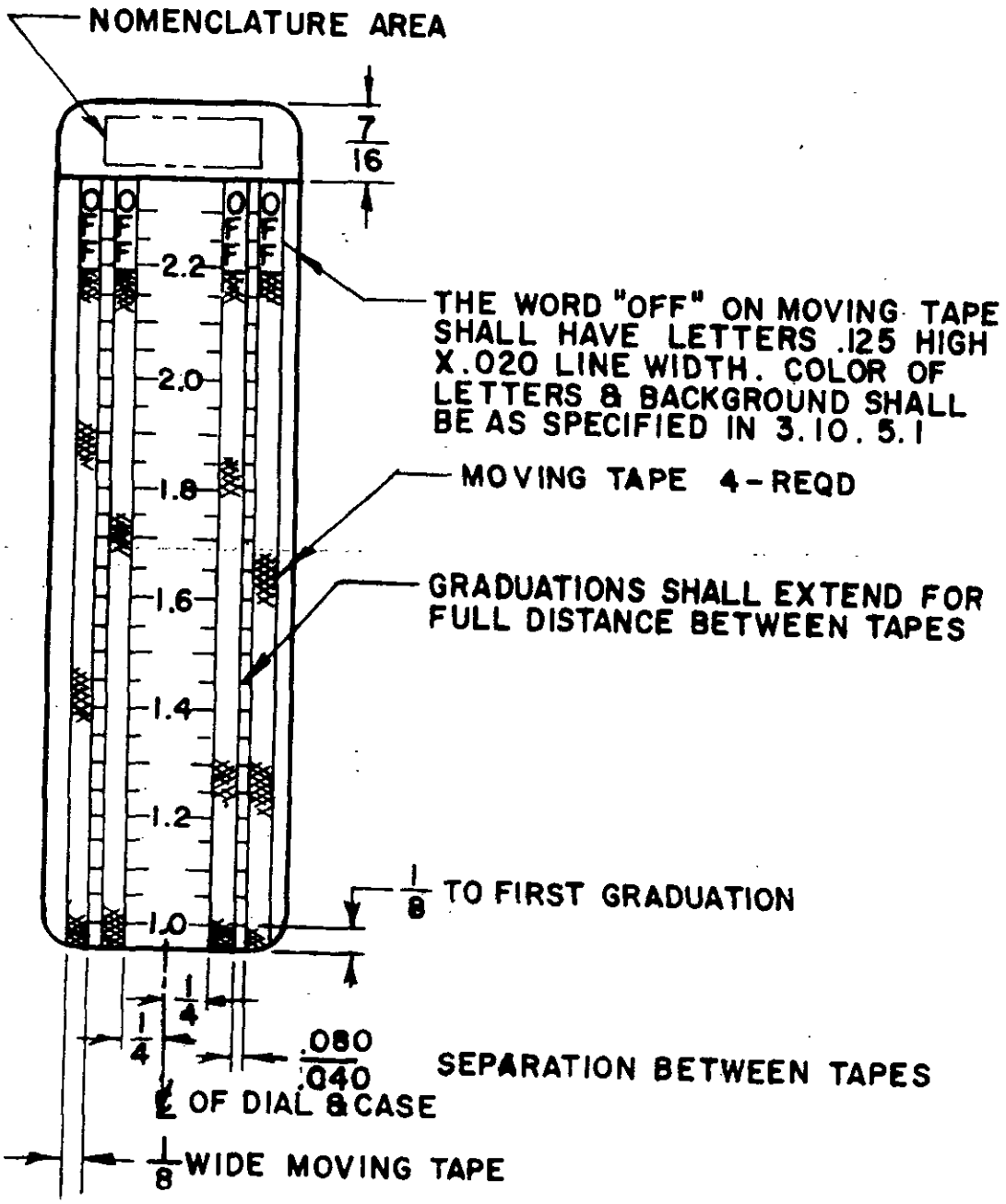
3.7.2.5 Dial. The indicator dial and cover glass shall be as shown on figure 4.

3.7.2.5.1 Scale. The scale shall conform to figure 5. The scale face shall be securely attached to the frame of the indicator mechanism.

3.7.2.5.2 Vertical tapes. The four vertical tapes shall be as shown on figure 4.

3.7.2.5.2.1 Tape drive stops. The drive mechanism for each respective indicator tape shall incorporate lower and upper stops that will limit tape travel from "Power-off Warning" to 2.5 engine pressure ratio units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:41Z
Check the source to verify that this is the current version before use.



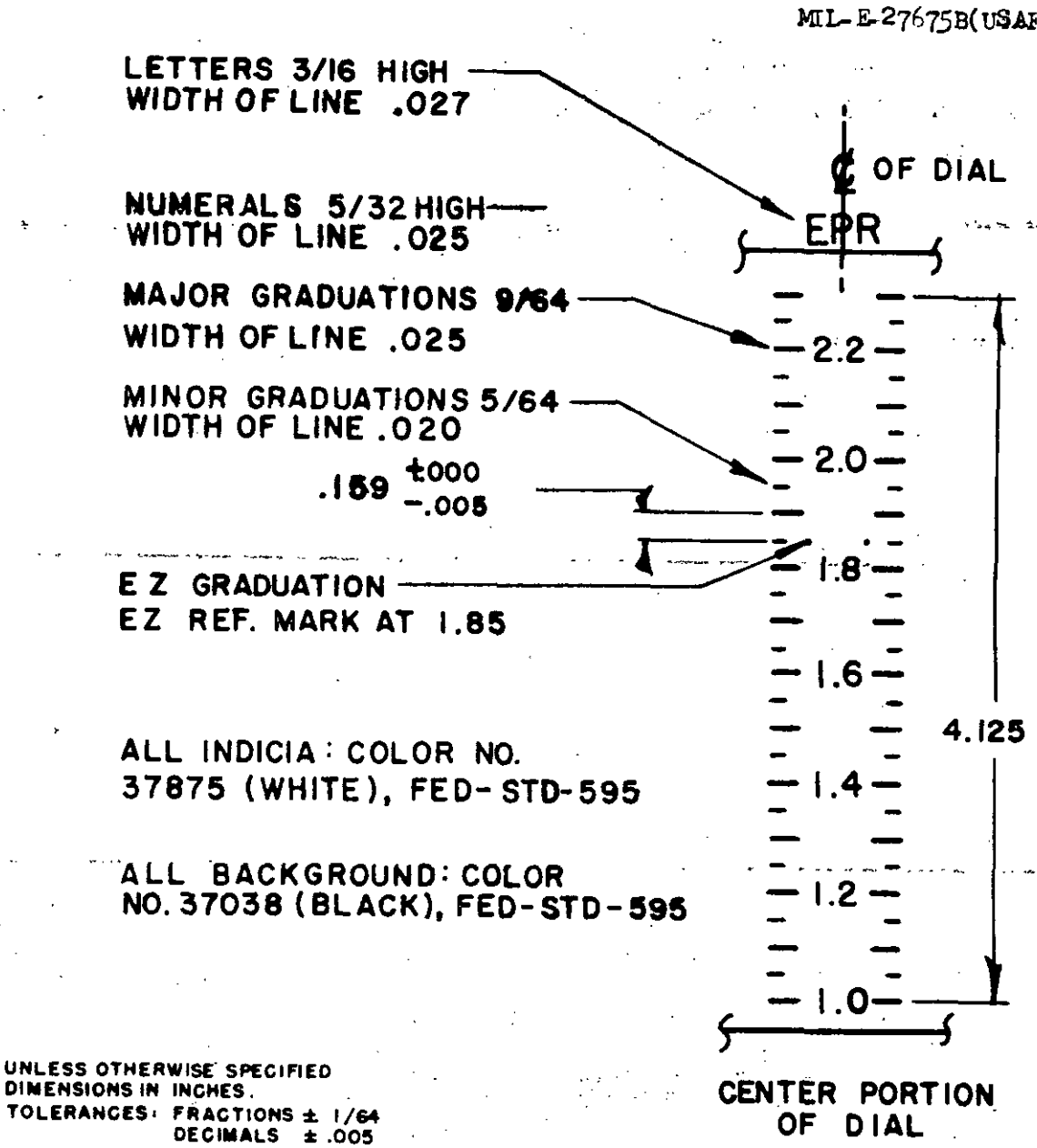
NOTES

WHITE PORTION OF MOVING TAPE SHALL BE COLOR NO. 37875 FED-STD-595.

SEE FIGURE 5 FOR SIZES OF GRADUATIONS, & NUMERALS; AND COLOR OF NUMERALS & BACKGROUND.

FIGURE 4 INDICATOR DIAL FACE

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:41Z
Check the source to verify that this is the current version before use.



UNLESS OTHERWISE SPECIFIED
DIMENSIONS IN INCHES.
TOLERANCES: FRACTIONS ± 1/64
DECIMALS ± .005

FIGURE 5 DIAL FACE DETAILS

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:41Z
Check the source to verify that this is the current version before use.

3.7.2.5.2.2 Off-scale indication. Since tape travel above the highest visible graduation will exist, the bottom (white) portion of the tape shall be equipped with a series of 3/64 inch diameter black dots to indicate movement of the tape. The distance between the center of the dots shall be equal to the distance between the lowest increments of scale graduations. The position of the highest dot on the tape shall be such that it will be at the lowest scale graduation when the top of the white portion of the tape is even with the highest scale graduation (dots will not be visible when the tapes are at normal operating positions). Sufficient dots shall be incorporated to permit indication to 2.5 EPR as they are read against the low scale graduation.

3.7.2.5.2.3 Power-off warning. Each EPR tape shall incorporate a power-off warning as shown on figure 4. When no input power is applied to a particular engine display, the tape shall display the word OFF. The warning shall disappear from view within 1 second after input power is applied.

3.7.2.5.3 Cover glass distance. The distance between the scale face and the inside of the cover glass shall be the minimum practicable and shall not exceed 0.100 inch.

3.7.2.5.4 Lighting. The indicator dial shall be integrally lighted in accordance with MIL-L-27160. Lighting acceptability shall be based on a visual comparison with a prime standard indicator.

3.7.2.5.4.1 Prime standard. A minimum of 20 areas covering the entire face of the standard indicator shall be measured for brightness level in foot-lamberts. The prime standard instrument and its recorded over-all brightness measurements shall be approved by the procuring activity. The standard shall be checked periodically, at least every 100 hours of operation, to determine any change in brightness. Variations greater that ±20 percent in two or more areas shall require return of the standard for corrective action. If a tape instrument is used as a prime standard, the tapes shall be fixed and shall be positioned in such a manner that a full scale reading and an OFF scale reading are represented.

3.7.2.5.5 Visibility. All scale markings and the total width of all tapes shall be visible from any position in front on the dial ranging between normal and a position 45° to the normal.

3.7.2.6 Electrical zero. The indicator synchro control transformers shall be so oriented that when the indicator is connected in accordance with figure 6 and 3.7.1.4 to a transmitter calibrated as specified in 3.7.1.9, the indicator shall display an EPR reading of 1.85 for a corresponding transmitter setting of electrical zero. In addition, when increasing pressure ratio values are sensed by the transmitter, the output signals produced shall cause the respective indicator to likewise register compatible increasing readings. The same relationship shall apply for decreasing pressure ratio measurements.

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:41Z
Check the source to verify that this is the currentversion before use.

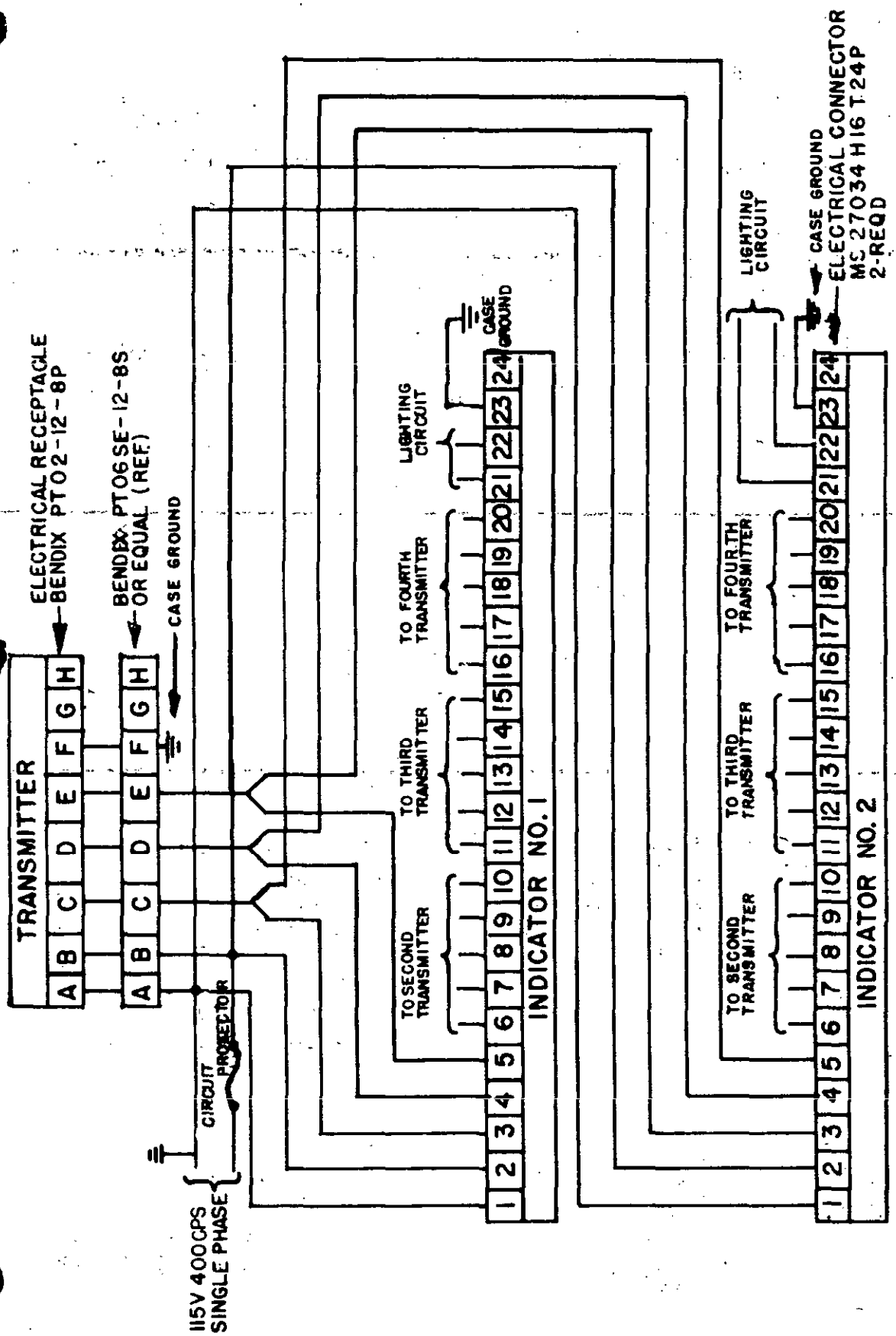


FIGURE 6 INTERCONNECTING WIRING DIAGRAM

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:41Z
Check the source to verify that this is the current version before use.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-70327 shall govern changes in the manufacturer's part numbers.

3.9 Radio noise suppression. The system radio noise suppression requirements shall be in accordance with MIL-I-6181 for the frequency range of 0.15 to 1,000 mc. Where filtering is required, the volume and weight of filtering components shall be minimized by proper electrical and mechanical design. The system shall not be susceptible to electrical interference exceeding the limits defined in MIL-I-6181.

3.10 Finishes, colors, and protective coatings

3.10.1 Steel parts. Steel parts, other than corrosion-resistant steel and not in moving contact, shall be cadmium plated in accordance with QQ-P-416, type II. Stainless steel parts shall be passivated.

3.10.2 Brass, bronze, and copper-alloy parts. Brass, bronze, and copper-alloy parts shall be plated or adequately treated for corrosion prevention as approved by the procuring activity.

3.10.3 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows:

a. Dials, small holes, and case inserts need not be anodized.

b. Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

c. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

d. When the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

e. Parts in a hermetically sealed container need not be anodized.

f. When the part is plated with tin over a copper flash, the part need not be anodized.

3.10.4 Transmitter case. The inside and outside surfaces of the case shall be suitably protected as specified in 3.10.3. The outside color of the case shall be natural aluminum. The inside surfaces of the case may be finished in durable dull black to provide improved heat - transfer characteristics.

3.10.4.1 Mounting rack. The aluminum mounting rack shall be suitably protected in accordance with 3.10.3. The color shall be natural aluminum.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:41Z
Check the source to verify that this is the current version before use.

3.10.5 Indicator. The indicator case, the background of the indicator dials, mounting bezels, gaskets, spacer rings, and non-indicating portions of the tape shall be finished in durable dull black, color number 37038 of FED-STD-595.

3.10.5.1 Power-off warning. The word "OFF" on the tapes shall be black, color number 17038 of FED-STD-595, on a red background. The background color shall be red fluorescent number 116-1548 (Switzer Brothers, Cleveland, Ohio), or equal.

3.10.6 Anti-reflection coating. All reflecting glass surfaces shall be provided with a reflection-reducing coating that withstands the environmental conditions specified herein and meets the requirements of MIL-C-675, except that the following reflectance tolerances shall apply:

Angle of Incidence	Wave Length Millimicrons	Average Percent Reflectance
0°	450 to 675	0.6
0°	425 to 700	0.5
30°	450 to 625	1.0
30°	425 to 700	0.5

Coated surfaces shall not chip, scale, peel, flake dissolve, discolor, or otherwise be adversely affected in normal service use. The coating on the outer surface of the coverglass shall withstand abrasion incident to normal service usage.

3.11 Operational markings

3.11.1 Indicator dial. Dial numbers and letters shall conform to MS33558.

3.11.1.1 Scale. The scale graduations and dimensions of the markings shall be in accordance with figure 5.

3.11.2 Wiring diagram. The external wiring diagram (transmitter to indicator) shall be permanently marked on a metal plate which shall be attached to the transmitter by means of screws or rivets. The wiring diagram shall be in accordance with figure 6.

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.12.1 Pressure connections. Pressure connection identification shall be permanently and legibly marked as shown on figure 1.

3.13 Workmanship. The transmitter and indicator shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring,

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:41Z
Check the source to verify that this is the current version before use.

marking of parts and assemblies, plating, painting, riveting, machine screw assembly, welding, brazing, and freedom of parts from burrs and sharp edges.

3.13.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with best shop practices. Where dimensions and tolerances may effect the interchangeability, operation, or performance of the system, they shall be held or limited accordingly.

3.13.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt threads.

3.13.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.13.4 Cleaning. The transmitter and indicator shall be thoroughly cleaned of loose, spattered or excess solder, metal chips or other foreign material after assembly. Burrs and sharp edges, as well as resin flash which might crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the engine pressure ratio measuring system shall be classified as follows:

a. Qualification tests.

b. Acceptance tests.

4.3 Test conditions

4.3.1 Atmospheric. Unless otherwise specified, all tests shall be conducted at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25° C). When tests are made with atmospheric pressures or temperatures substantially different from the above values, proper allowance shall be made from the specified condition.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:41Z
Check the source to verify that this is the current version before use.

4.3.2 Attitude. Unless otherwise specified, the system shall be tested in normal operating position.

4.3.3 Test indicator. When a pressure ratio indicator is specified for testing, it shall meet the design, performance, and accuracy requirements specified herein.

4.4 Qualification testing

4.4.1 Test samples. The test samples shall consist of three transmitters and three indicators representative of the production equipment. Samples shall be identified with the manufacturer's part number and any other information as required by the procuring activity.

4.4.2 Test report. After the contractor completes the qualification tests, he shall prepare a test report in accordance with MIL-STD-831 and furnish six complete copies to the procuring activity.

4.4.3 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6.

4.4.3.1 Qualification test schedule. The qualification schedule shall be as specified in table III for the transmitters and table IV for the indicators.

4.4.3.2 Fault analysis. A detailed fault analysis showing compliance with the requirements specified herein shall be prepared by the contractor and submitted to the procuring activity. The analysis shall show the types and probabilities of failures and the effects of operation loss of various components on the over-all system operation.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests
b. Sampling plans and tests.

4.5.1 Individual tests. Each transmitter and indicator shall be subjected to the following tests as described under 4.6.

4.5.1.1 Transmitter

a. Examination of product
b. Leakage
c. Short scale error at room temperature
d. Position error
e. Friction error

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:41Z
Check the source to verify that this is the current version before use.

TABLE III. Transmitter Test Schedule

Paragraph No.	Test	Test Sample #1	Test Sample #2	Test Sample #3
4.6.1.1	Examination of product	x	x	x
4.6.1.2	Leakage	x	x	x
4.6.1.3	Scale error at room temperature	x	x	x
4.6.1.4	Position error	x	x	x
4.6.1.5	Friction error	x	x	x
4.6.1.6	Response rate	x	x	x
4.6.1.7	Sensitivity	x	x	x
4.6.1.8	Overpressure	x	x	x
4.6.1.9	Dielectric strength			x
4.6.1.10	Radio noise		x	
4.6.1.11	Low temperature	x	x	x
4.6.1.12	High temperature	x	x	x
4.6.1.13	Extreme low temperature exposure			x
4.6.1.14	Extreme high temperature exposure			x
4.6.1.15	Vibration, resonance, scale error after vibration			x
4.6.1.16	Humidity		x	
4.6.1.17	Fungus			x
4.6.1.18	Salt spray		x	
4.6.1.19	Sand and dust			x
4.6.1.20	Sealing			x
4.6.1.21	Endurance and reliability	x		
4.6.3.1	Compatibility		x	
4.6.3.2	Maintainability		x	

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:41Z
Check the source to verify that this is the current version before use.

TABLE IV. Indicator Test Schedule

Paragraph No.	Test	Test Sample #1	Test Sample #2	Test Sample #3
4.6.2.1	Examination of product	x	x	x
4.6.2.2	Electrical zero	x	x	x
4.6.2.3	Scale error at room temperature	x	x	x
4.6.2.3.1	Position error	x	x	x
4.6.2.4	Friction error	x	x	x
4.6.2.5	Vibration			x
4.6.2.6	Low temperature	x	x	x
4.6.2.7	High temperature	x	x	x
4.6.2.8	Magnetic effect		x	
4.6.2.9	Voltage and frequency variation			x
4.6.2.10	Sealing	x		
4.6.2.11	Indicator response	x	x	x
4.6.2.12	Indicator sensitivity	x	x	x
4.6.4.13	Extreme low temperature exposure			x
4.6.2.14	High temperature exposure			x
4.6.2.15	Thermal Shock cycling	x	x	x
4.6.2.16	Integral lighting	x	x	x
4.6.2.17	Salt spray	x		
4.6.2.18	Endurance and reliability	x		
4.6.2.19	Radio noise interference			x
4.6.2.20	Dielectric strength	x		
4.6.2.21	Phasing-Indicator to tranamitter	x	x	x
4.6.2.22	Decompression		x	
4.6.2.23	External Pressure		x	
4.6.3.1	Compatibility		x	
4.6.3.2	Maintainability		x	

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:41Z
Check the source to verify that this is the current version before use.

4.5.1.2 Indicator

a. Examination of product
b. Electrical zero
c. Scale error at room temperature
d. Friction error at room temperature
e. Position error
f. Indicator response
g. Sensitivity
h. Integral lighting (visual comparison)
i. Phasing - Indicator to transmitter

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. Three transmitters and three indicators shall be selected at random from each 100 or fraction thereof produced and subjected to the following tests as described under 4.6.

4.5.2.1.1 Transmitter

a. Response rate
b. Sensitivity
c. Overpressure
d. Dielectric strength
e. Radio noise
f. Low temperature
g. High temperature

4.5.2.1.2 Indicator

a. Vibration
b. Scale error at low temperature
c. Scale error at room temperature
d. Scale error at high temperature
e. Magnetic effect
f. Voltage and frequency variation
g. Sealing
h. Thermal shock cycling
i. Dielectric strength
j. Integral lighting (Sampling Plan A of MIL-L-27160)

4.5.2.2 Sampling plan B. Unless otherwise specified, three transmitters and three indicators shall be selected at random from the first 10 items produced and subjected to the following tests as described under 4.6:

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:41Z
Check the source to verify that this is the current version before use.

4.5.2.2.1 Transmitter

a. Sampling plan A tests
b. Scale error at room temperature, long
c. Extreme low temperature exposure
d. Extreme high temperature exposure
e. Vibration
f. Humidity
g. Fungus
h. Salt spray
i. Sand and dust
j. Sealing
k. Endurance and reliability
l. Compatibility
m. Maintainability

4.5.2.2.2 Indicator

a. Sampling plan A tests
b. Extreme low temperature exposure
c. High temperature exposure
d. Salt spray
e. Endurance
f. Radio noise suppression
g. Integral lighting (Sampling plan B of MIL-L-27160)
h. Decompression
i. External pressure
j. Compatibility
k. Maintainability

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual test may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:41Z
Check the source to verify that this is the current version before use.

4.6 Test methods

4.6.1 Transmitter tests

4.6.1.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.1.2 Leakage. A pressure of 42 inches Hg absolute shall be applied to the P_t2 port. The pressure line shall then be sealed upstream of the manometer and with the P_t7 port open to ambient pressure, the P_t2 pressure shall not decrease more than 0.05 inch Hg in 5 minutes. A pressure of 96 inches Hg absolute shall be applied to the P_t7 port. The pressure line shall then be sealed upstream of the manometer and with the P_t2 port held at approximately 35 inches Hg absolute, the P_t7 pressure shall not decrease morethan 0.05 inch in 5 minutes. The internal volume of the isolated part of the system (manometer wells, pressure lines, et cetera) shall not exceed 45 cubic inches.

4.6.1.3 Scale error at room temperature. The transmitter shall be connected to a suitable indicator having a known accuracy and a load equivalent to two tape indicators and subjected to a scale error at room temperature test at the test points specified in table V. The tolerances shall not exceed those specified. The temperature shall be maintained within 25° ±2° C. During this test, the transmitter shall be vibrated at a frequency of 60 cps with an amplitude not exceeding 0.002 inch. This test shall be conducted both up scale and down scale.

4.6.1.3.1 Short scale error. When a short error test is specified, the test points shall be those indicated as E in table V. Reading before tapping need not be taken unless a friction error test is specified in conjunction with this test.

4.6.1.4 Position error. A position error test shall be conducted at 0°, 45°, 135°, 225°, and 315° around the major and minor axes. The transmitter shall be subjected to this test only at room temperature and at the test points indicated as P in table V. Position error tolerance at the 45° and 315° positions around the major and minor axes shall not exceed ±0.005 EPR. Position error tolerance at the 135° and 225° positions around the major and minor axes shall not exceed ±0.015 EPR.

4.6.1.5 Friction error. A friction error test shall be conducted at the test points designated E in table V. The tolerances shall not exceed those specified in table V.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:41Z
Check the source to verify that this is the current version before use.

TABLE V

Engine Inlet Pressure (In. Hg Abs)	Engine Exhaust Pressure (Inches Hg Absolute)								
3.00	3.00	3.30	3.60	EP 4.20	4.80	S 5.40	6.00		
4.00				5.60	6.40	E 7.20	8.00		
7.50		8.25	9.00	EP 10.50	12.00	E 13.50	15.00	E 16.50	EP 17.25
11.00	11.00			15.40	EP 17.60	19.80	S 22.00		25.30
15.00		16.50	18.00	21.00	24.00	EP 27.00	30.00	33.00	34.50
20.00		22.00	24.00	28.00	E 32.00	36.00	40.00	EP 44.00	46.00
25.00	E 25.00	27.50	E 30.00	35.00	E 40.00	45.00	E 50.00	55.00	
28.00		EP 30.80	33.60	39.20	EP 44.80	50.40	56.00	E 61.60	
35.00	35.00	38.50	42.00	49.00	56.00	E 63.00	70.00	EP 77.00	80.50
42.00		46.20	50.40	58.80	67.20	S 75.60	E 84.00	92.40	96.60
Pressure Ratio	1.00	1.10	1.20	1.40	1.60	1.80	2.00	2.20	2.30

P - Position error test points
E - Short scale error test points
S - Sensitivity test points
Scale Error Tolerances for Room Temperature: 1.0 to 1.5 EPR = ±0.020 EPR
1.5 thru 2.3 EPR = ±0.015 EPR
Scale Error Tolerances for Low and High Temperatures, Vibration, and Salt Spray: 1.0 to 1.5 EPR = ±0.025 EPR
1.5 thru 2.3 EPR = ±0.020 EPR
Scale Error Tolerances After Endurance: 1.0 to 1.5 EPR = ±0.030 EPR
1.5 thru 2.3 EPR = ±0.025 EPR
Friction Error Tolerances for Room, Low and High Temperatures:
1.0 thru 2.3 EPR = ±0.005 EPR
Position Error Tolerances for Room Temperatures:
At positions 45° and 315° from horizontal, 1.0 thru 2.3 EPR = ±0.005 EPR
At positions 135° and 225° from horizontal, 1.0 thru 2.3 EPR = ±0.015 EPR

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:41Z
Check the source to verify that this is the current version before use.

4.6.1.6 Response rate. Pressure shall be applied to the engine inlet pressure connection per table I and stabilized to each test point. Pressure then shall be applied to the engine exhaust pressure connection per the value listed in table I corresponding to the engine inlet pressure used. When the pressures are stabilized, the pressure shall be so blocked that the pressures in the tubing remain the same as those in the transmitter. The power shall remain on during this test. The quick acting valve shall be closed, and pressure shall then be raised to and stabilized at the second pressure in table I corresponding to the engine inlet pressure used. The quick acting valve shall be opened. The response time during each step tested shall not exceed the times specified in table I. The transmitter shall be considered stabilized when the output variation does not exceed 0.003 pressure ratio unit. The transmitter output shall begin to vary within 0.25 second after application of power to the quick acting valve. The transmitter output shall stabilize within one and one-half seconds of the time the output first passes the final reading. This time is not in addition to the total response time. This procedure shall be repeated at the same test points for decreasing engine exhaust pressure.

4.6.1.7 Sensitivity. The engine inlet pressure and the engine exhaust pressure shall be adjusted and stabilized to the pressures indicated in table II. While holding the engine inlet pressure constant, the engine exhaust pressure shall then be increased until a change of indication is first detected. The exhaust pressure shall then be decreased slowly until a change of indication is detected. The difference in readings shall not exceed the values specified in table II. The procedure shall be repeated by varying the inlet pressure and holding the engine exhaust pressure constant. A suitable indicator of known sensitivity shall be used for this test.

4.6.1.8 Over pressure. The transmitter shall be subjected to a P_t2 pressure of 93.5 ±1 inch Hg absolute and P_t7 pressure of 165 ±1 inch Hg absolute. The resultant indicated pressure ratio shall be accurate to within ±0.100 pressure ratio units. The short scale error at room temperature test shall then be conducted and the errors shall not exceed those specified in table V.

4.6.1.9 Dielectric strength. The transmitter shall be tested for dielectric strength. A voltage of 1,250 volts ac rms at commercial frequency shall be applied for one minute to all parts of nonhermetically sealed transmitters that carry 65 to 115 volts. A voltage of 500 volts ac rms at commercial frequency shall be applied for one minute to all part of nonhermetically sealed transmitters that carry 65 volts or less. Transistors and similar sensitive parts may be disconnected from the circuit prior to testing. A voltage of 200 volts ac rms at commercial frequency shall be applied to hermetically sealed transmitters for one minute. If hermetically sealed transmitters are unsealed to disconnect transistors or sensitive parts, they need not be resealed but shall be tested as nonhermetically sealed transmetters.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:41Z
Check the source to verify that this is the current version before use.

4.6.1.10 Radio noise. The transmitter shall be subjected to transmitted and susceptibility noise suppression tests in accordance with MIL-I-6181.

4.6.1.11 Low temperature. The transmitter shall be placed in a test chamber and subjected to a temperature of -54° ±2° C for a period of 4 hours. While at this temperature, the short scale error test shall be conducted (see 4.6.1.3.1). The scale error tolerances shall not exceed those specified in table V. At the completion of this test, the transmitter shall be subjected to a short scale error at room temperature test. The tolerances shall not exceed those specified.

4.6.1.12 High temperature. The transmitter shall be placed in a test chamber and subjected to a temperature of 100° ±2° C for a period of 4 hours. While at this temperature, the short scale error test shall be conducted (see 4.6.1.3.1). The scale error tolerances shall not exceed those specified in table V. The transmitter shall then be subjected to a short scale error at room temperature test. The tolerances shall not exceed those specified.

4.6.1.13 Extreme low temperature exposure. The transmitter shall be placed in a test chamber and subjected to a temperature of -65° ±2° C for a period of 48 hours. The temperature shall then be raised to -54° ±2° C and maintained for 24 hours. While still at -54° ±2° C, the transmitter shall be subjected to a short scale error test. Scale error tolerances shall not exceed those specified. The transmitter shall be returned to room temperature for a minimum period of 4 hours and subjected to the scale error at room temperature.

4.6.1.14 Extreme high temperature exposure. The transmitter shall be placed in a test chamber and subjected to a temperature of 121° ±2° C for a period of 24 hours. Power shall be applied continuously to the transmitter throughout the last 8 hours of this period. While still at 121° ±2° C, the transmitter shall be operated throughout its range and shall exhibit no sticking or erratic operation. The transmitter shall be returned to room temperature for a minimum period of 4 hours and subjected to a short scale error at room temperature test.

4.6.1.15 Vibration. The transmitter shall be tested within the range of frequencies and vibratory accelerations specified in MIL-E-5272, procedure XII, curve A.

4.6.1.15.1 Resonant frequencies. Function of the transmitter shall be checked concurrently with the operation of scanning the frequency range for resonant frequency. The transmitter shall be vibrated for 1 hour through each of three mutually perpendicular axes of vibration at the resonant condition. When more than one resonant frequency is encountered with vibration applied along any one axis, the test shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:41Z
Check the source to verify that this is the current version before use.

be accomplished at the most severe resonance or the test period may be divided among the resonant frequencies, whichever is most likely to product failure. This test shall be conducted at room temperature.

4.6.1.15.1.1 When resonant frequencies are not apparent within the specified frequency range, the transmitter shall be vibrated for twice the specified time period at a frequency of 55 cps and an applied double amplitude of 0.060 inch.

4.6.1.15.2 Scale error after vibration. Following the vibration period, the transmitter shall be tested in accordance with 4.6.1.3.1. Allowable tolerances specified in table V shall apply.

4.6.1.16 Humidity. Except when completely hermetically sealed, the transmitter shall be subjected to the humidity test in accordance with procedure I of MIL-E-5272.

4.6.1.17 Fungus. Except when completely hermetically sealed, the transmitter shall be subjected to the fungus test in accordance with procedure I of MIL-E-5272 for a period of 14 days.

4.6.1.18 Salt spray. The transmitter shall be subjected to the salt spray test in accordance with procedure I of MIL-E-5272. Upon completion of the test, the transmitter shall be subjected to the scale error at room temperature test specified in 4.6.1.3.1. The scale errors shall not exceed the room temperature tolerances specified in table V.

4.6.1.19 Sand and dust. The transmitter shall be subjected to a sand and dust test in accordance with procedure I of MIL-E-5272, and then subjected to a short scale error at room temperature test. The tolerances shall not exceed those specified in table V.

4.6.1.20 Sealing. If hermetically sealed, the transmitter shall be tested by means of a mass spectrometer type of helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 1.0 micron cubic foot per hour at a pressure differential of approximately 1 atmosphere. Where 100 percent helium is utilized as a filling medium, the leak rate shall not exceed 10.0 micron cubic foot per hour.

4.6.1.21 Endurance and reliability. The transmitter shall be subjected to an endurance and reliability test in accordance with the following:

MAX. ALLOWABLE FAILURES	TOTAL TEST TIME (HOURS)
0	1726
1	2250
2	2685
3	3200
4	3720

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:41Z
Check the source to verify that this is the current version before use.

GPO 813-342-3

The transmitter shall be tested for at least 1726 hours of actual operation. Eight hundred and sixty three (863) hours shall be accomplished while P_t2 is varied from ambient to 11.5, +0.5 or -0.0 inch Hg absolute and return to ambient at a rate of 5 ±2 cpm. Upon completion of each 100 hours of operation, P_t2 shall be varied once from ambient to 11, +0.0 or -0.5 inch Hg absolute and return to ambient within 10 seconds. During the other 863 hours, P_t7 shall be varied from ambient to 80, +0.0 or -0.5 inch Hg absolute and return to ambient at a rate of 5 ±2 cpm. Upon completion of each 100 hours of operation, P_t7 shall be varied once from ambient to 82, +0.5 or -0.0 inch Hg absolute and return to ambient within 10 seconds. The respective pressure sensing elements not being cycled shall be vented to ambient. The transmitter shall be subjected to a short scale error test upon completion of each 250 hours of P_t2 and P_t7 cycling operation. The tolerances specified in table V shall apply. No recalibration shall be required to pass the error test. In case of malfunction, the malfunction shall be recorded and the test interrupted while repairs are made. The test shall then be continued from the point of interruption. This test may be interrupted for normal week-end shutdowns. Satisfactory completion of this test (the maximum failures shall not exceed those listed above) will demonstrate with 90 percent confidence that MTBF of at least 750 hours has been achieved.

4.6.2 Indicator

4.6.2.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2.2 Electrical zero. The indicator electrical zero shall be checked in accordance with procedures outlined in MIL-S-20708. Upon application of power, the top of the white tape column shall align itself with the longitudinal center of the electrical zero reference mark (±0.005 inch) shown on figure 5. The procedure shall be repeated until all four indicator tapes have been checked for electrical zero.

4.6.2.3 Scale error at room temperature. The input signal values applicable to the test points specified in table VI shall be applied to the indicator by means of a master synchro transmitter having a known accuracy. The scale errors of the indicator shall be noted at the specified test points and the tolerances shall not exceed those specified in table VI.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:41Z
Check the source to verify that this is the current version before use.

TABLE VI. Indicator Pressure Ratio Tolerances

Pressure Ratio	Tolerance ±Pressure Ratio		
	High Temperature	Low Temperature	Room Temperature
1.0	0.005	0.005	0.004
1.2	0.005	0.005	0.004
1.4	0.005	0.005	0.004
1.8	0.005	0.005	0.004
2.0	0.005	0.005	0.004
2.3	0.005	0.005	0.004

4.6.2.3.1 Position error. The input signal value shall be varied to drive the indicator tape to midscale indication and locked. The reading of the indicator taken while in a normal upright position shall not differ from the readings obtained when the indicator is held at 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° around the major and minor axes by more than the allowable tolerances specified in table VII.

TABLE VII

Test	Tolerance ±Pressure Ratio
Frequency and Voltage Variation	0.005
Position Error	0.005
Friction:	
Room Temperature	0.005
Low Temperature	0.010
High Temperature	0.010

4.6.2.4 Friction error. The friction error test may be conducted in conjunction with the scale error at room temperature test at the test points specified in table VI. After a slow change in increasing indication, the difference in indication shall be noted before and after the indicator is lightly tapped. The procedure shall be repeated for decreasing indications. The maximum observed difference in indication shall not exceed the tolerance allowed for friction in table VII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:41Z
Check the source to verify that this is the current version before use.

4.6.2.5 Vibration. The indicators shall be subjected to the "Supplementary test, electronic assemblies" portion only of the vibration test specified in MIL-E-5272, procedure XII. During cycling, each indicator shall be so energized that its normal tape position will indicate 1.9 EPR unit. At no time shall tape oscillation or position deviate from 1.9 EPR unit by more than 0.005 EPR unit. No condition of applied vibration shall cause any apparent change in the OFF conditions. Any condition that would result in false indication of a warning position will be unacceptable. No malfunction of the indicator shall result from this test.

4.6.2.5.1 Scale error after vibration. Following the vibration test, the indicators shall be checked for scale error at room temperature and friction error. The tolerances specified in tables VI and VII shall apply.

4.6.2.6 Low temperature. The indicator shall be placed in a test chamber and subjected to -54° ±2° C for a period of 4 hours. At the end of this period and while still at the specified temperature, the indicator shall be subjected to the test specified in 4.6.2.3. The indicator scale error shall not exceed the tolerance specified in table VI. The indicator shall be returned to room temperature for a minimum period of 4 hours and again subjected to the scale error at room temperature test. The scale error shall not exceed the tolerance specified in table VI.

4.6.2.7 High temperature. The indicator shall be placed in a test chamber and subjected to a temperature of +71° ±2°C for a period of 4 hours. At the end of this period and while still at this temperature, the indicator shall be subjected to the test specified in 4.6.2.3. The scale error shall not exceed the tolerance specified in table VI. The indicator shall be returned to room temperature for a minimum period of 4 hours and again subjected to the scale error at room temperature test. The scale error shall not exceed the tolerance specified in table VI.

4.6.2.8 Magnetic effect. The indicator, operating on rated voltage shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5 ±1/2 inch from, and magnetically east or west of, the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated about its own horizontal axis until it is in its normal viewing position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 5 degrees.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:41Z
Check the source to verify that this is the current version before use.

4.6.2.9 Voltage and frequency variation. The indicator shall be subjected to the test specified in 4.6.2.3 and the input power varied to provide any combination of the following voltages and frequencies. The scale errors of the indicator at the varied voltages and frequencies shall not exceed the tolerances specified in table VII.

a. 104V to 126V

b. 380 to 420 cps

4.6.2.10 Sealing. The hermetically sealed indicator shall be tested for leaks by means of a mass spectrometer type of helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 1.0 micron cubic foot per hour at a pressure differential of approximately 1 atmosphere. Where 100 percent helium is utilized as the filling medium, the leak rate shall not exceed 10.0 micron cubic foot per hour.

4.6.2.11 Indicator response. This test shall be conducted by simulating input signals to the indicator to indicate 1.0 EPR on the scale. The input signals shall then be instantaneously increased to values that will produce full scale indication (2.3 EPR). The time required to reach full scale deflection shall not exceed 6 seconds.

4.6.2.12 Indicator sensitivity. Input signals shall be provided to the indicator which will drive the tape indication to 1.5 EPR. The input signal shall be slowly varied to equal a 0.05 EPR indication change. The resultant indicator readings shall change with 0.05 ±0.01 EPR. This procedure shall be repeated at the 1.8 and 2.2 EPR test points.

4.6.2.13 Extreme low temperature exposure. The indicator shall be placed in a test chamber and subjected to a temperature of -65° ±2° C for a period of 48 hours. While at this temperature, no operational test shall be performed. The temperature shall be raised to -54° ±2° C, and maintained for 24 hours. While still at -54° ±2° C, the indicator shall be subjected to scale error at the test points specified in table VI. A scale error tolerance of ±0.005 EPR shall apply. The indicator shall be returned to room temperature for a minimum period of 4 hours and subjected to the test specified in 4.6.2.3.

4.6.2.14 High temperature exposure. The indicator shall be placed in a chamber and subjected to a temperature of 71° ±2° C for a period of 24 hours. Normal instrumentation power shall be applied continuously the last 2 hours of this period. While still at 71° ±2° C, the indicator shall be operated throughout its range and shall exhibit no sticking or erratic characteristics. The indicator shall then be subjected to and shall meet

Source: https://assist.dla.mil -- Downloaded: 20261010-20T12:41Z
Check the source to verify that this is the current version before use.

the tolerances specified in table VI. The indicator shall be returned to room temperature for a minimum period of 2 hours and subjected to the scale error at room temperature test. The scale errors shall not exceed the tolerances specified in table VI. The indicators shall then be subjected to the response test. The response time shall not exceed 6 seconds.

4.6.2.15 Thermal shock cycling. The indicator shall be subjected to a total of 8 cycles of immersion in tap water at a temperature of 85° ±2° C and then in tap water at a temperature of 5° ±2° C. The length of time for each bath immersion shall be 30 minutes with not more than 3 seconds between immersions. No evidence of moisture penetration or damage to the case shall result from this test. Following this test, the indicators shall be subjected to and shall meet the test specified in 4.6.2.10. Fogging of the cover glass during any part of this test shall be cause for rejection.

4.6.2.16 Integral lighting. Integral lighting tests shall be conducted in accordance with MIL-L-27160.

4.6.2.16.1 Visual comparison. Indicator lighting shall be checked by comparison with a prime standard. The prime standard and the instrument under test shall be placed side by side and at eye level. The brightness of the two units shall be compared with 2.85V ±0.1V applied to both the standard and the unit under test. General brightness lighting level comparison shall then be made between the indicator being tested and the prime standard, with the prime standard voltage at 2.7V and then at 3.0V. If the general brightness level falls within the brightness limits of the prime standard when energized at the specific voltage levels, the indicator shall be acceptable. If any area of the instrument under test appears visually to differ from the same area of the prime standard, it shall then be subjected to the brightness tests specified in MIL-L-27160, and sufficient data recorded to substantiate conformance with lighting requirements.

4.6.2.16.2 Color comparison. A color comparison shall be made by applying 4.5V to the prime standard and to the indicator under test. The two indicators shall be compared and shall visually appear identical in color. If the instrument under test appears to be a different color than the prime, it shall then be subjected to the color tests specified in MIL-L-27160.

4.6.2.17 Salt spray. The indicator shall be subjected to a salt spray test in accordance with MIL-E-5272, procedure I. The scale error at room temperature test shall be conducted immediately before and after this test. The scale error tolerances shall not exceed those specified for room temperature. Subsequent indicator operation shall not be affected by this test, and excessive corrosion of metal parts, including electrical receptacles, shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:41Z
Check the source to verify that this is the current version before use.

4.6.2.18 Endurance and reliability. The indicator shall be subjected to an endurance and reliability test in accordance with the following:

MAX. ALLOWABLE FAILURES	TOTAL TEST TIME (HOURS)
0	1726
1	2250
2	2685
3	3200
4	3720

The indicator shall be tested for at least 1726 hours of actual electrical operation. During this time simulated input signals shall be applied to the indicator to accomplish 12,000 cycles of approximately full scale deflection at a rate of 4 ±1 cycles per minute. The test shall begin when rated electrical power is applied. After completion of four hours of operation with rated power on, which includes 30 cycles, the power shall be turned off and remain off for a period of 30 minutes. The power-on, power-off procedure shall be repeated until the specified 1726 hours and 12,000 cycles of electrical operation have been achieved. In case of malfunction, the malfunction shall be recorded and the test interrupted while repairs are made. The test shall then be continued from the point of interruption. Upon completion of each 250 hours of cycling operation, the indicator shall be checked for scale error. The scale error readings shall not differ by more than 1.0 percent of full scale from those obtained at room temperature previous to the test. No recalibration shall be required to pass the error test. This test may be interrupted for normal weekend shutdowns. Satisfactory completion of this test (the maximum failures shall not exceed those listed above) will demonstrate with 90 percent confidence that MTBF of at least 750 hours has been achieved.

4.6.2.19 Radio noise. The indicator shall be subjected to transmitted and susceptibility noise suppression tests in accordance with MIL-I-6181. The conducted and radiated noise shall not exceed the values specified therein.

4.6.2.20 Dielectric strength. The indicator shall be tested for dielectric strength. A voltage of 200 volts ac rms at commercial frequency shall be applied to hermetically sealed indicators for one minute. If the indicators are unsealed to disconnect transistors or sensitive parts, they need not be resealed but shall be tested as nonhermetically sealed indicators in accordance with 4.6.1.9.

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:41Z
Check the source to verify that this is the current version before use.

4.6.2.21 Phasing - indicator to transmitter. The ability of each synchro receiving device in the indicator to seek the null position in a direction that will keep the indicator reading in phase with the transmitter synchro signal shall be demonstrated. This indicator to transmitter phasing shall be demonstrated by applying normal excitation power and setting the indicator tapes to read 2.0 EPR by manipulating the transmitter synchro. While holding this setting of the transmitter synchro, the excitation power shall be removed and the indicator tapes allowed to read "OFF". The excitation power shall then be reapplied and the reading of the indication tapes observed. Failure of the indicator tapes to return to 2.0 ±0.005 EPR reading within 6 seconds after application of excitation power shall be cause for rejection. After it has been determined that the tapes return to the 2.0 EPR reading, the excitation power shall again be removed and the indicator tapes allowed to read "OFF". Before reapplying excitation power, the transmitter synchro shall be rotated to a position equivalent to a reading of 1.0 EPR. The excitation power shall then be reapplied and the reading of the indicator tapes observed. Failure of the indicator tapes to exhibit a 1.0 EPR reading within 6 seconds after application of excitation power shall be cause for rejection. The above tests shall be repeated using initial tape settings of 2.3 and 2.5 EPR.

4.6.2.22 Decompression. The indicators shall be placed in a cabinet in which the pressure is maintained at the same pressure as inside the hermetically sealed case. The pressure within the cabinet shall be so released that a pressure differential of 16 inches Hg is instantaneously applied between the interior and the exterior of the case. Operation on the indicators shall not be affected by the sudden decrease in pressure.

4.6.2.23 External pressure. The indicator shall be subjected to a pressurization test by applying an absolute pressure of 56 inches Hg (approximately 26 inches Hg above atmospheric) for a period of 15 minutes with rated power on the indicator. Normal operation shall not be affected by this test and no failures of the cover glass or hermetic seal shall occur.

4.6.3 System tests

4.6.3.1 Compatibility. The TRK-55 pressure transmitter shall be connected to two channels of an EGK-7 indicator. Both units shall be connected to normal excitation power and ambient pressure applied to both the engine inlet pressure and engine exhaust pressure connections of the transmitter. The indicator tapes shall read 1.0 EPR. The pressure applied to the engine exhaust connection shall then be raised to 2.3 times the ambient pressure while maintaining ambient pressure at the engine inlet pressure connection. The indicator tapes shall read 2.3 EPR. The pressure

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:41Z
Check the source to verify that this is the current version before use.

applied to the engine exhaust connection shall then be lowered to ambient and maintained while the pressure applied to the engine inlet pressure connection is lowered to 0.4 times the ambient. The indicator tapes shall read 2.3 EPR. The above tests shall be repeated in a reverse direction. Failure of the indicator tapes to display the specified readings shall be cause for rejection. This test shall be repeated using the remaining two channels in the test indicator.

4.6.3.2 Maintainability. The system maintainability shall be verified in accordance with MIL-M-26512, Paragraph 40, Appendix B.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Levels A and C. Components of the system shall be preserved in accordance with method IA of MIL-P-116 except that when electrical connectors and pressure connections are equipped with water-vaporproof caps and gaskets or plugs, method III may be used. Packaging shall be in accordance with MIL-P-7936.

5.2 Packing

5.2.1 Levels A, B, and C. Components of the system shall be packed in accordance with the applicable levels of MIL-P-7936.

5.3 Marking for shipment. The shipment marking nomenclature shall be as follows:

"Engine Pressure Ratio System, Turbofan Engine A/A24J-13"

6. NOTES

6.1 Intended use. The A/A24J-13 engine pressure ratio system covered by this specification is intended for use in indicating aircraft turbofan engine pressure ratio measurements.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. If sampling plan B tests are to be conducted

c. Selection of applicable levels of packaging and packing.

GPO 813-342-2

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:41Z
Check the source to verify that this is the current version before use.

6.3 Definitions

6.3.1 Hermetic seal. Hermetic sealing is the process by which an item is totally enclosed by a suitable metal structure or case and sealed airtight by fusion of metallic or ceramic materials. This includes the fusion of metals by welding, brazing, or soldering; the fusion of ceramic materials under heat or pressure; and the fusion of ceramic materials into a metallic support. Elastomeric or resinous materials, or combinations of these materials, may be used as a coverglass seal provided the specified leak rate requirements are met under all tests and environmental conditions specified herein. Elastomeric or resinous materials may be used in other areas provided prior approval is obtained from the procuring activity. A hermetic seal is not intended to include seals accomplished by gaskets.

6.3.2 Amplitude. Amplitude is the extent of motion as measured from one extreme to the opposite extreme.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Systems Engineering Group, Attn: SELS, Wright-Patterson Air Force Base, Ohio and information pertaining to qualification of products may be obtained from that activity.

Custodian:
Air Force - 11

Preparing Activity:
Air Force - 11

Source: https://assist.dla.mil - Downloaded: 2026-10-20T12:41Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD Form 1426

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:41Z
Check the source to verify that this is the current version before use.